NETWORKING SEMINAR

Constraint-Based Routing and MPLS

Why Internet routing needs more than the shortest path, and how it is built

Younis & Fahmy (Purdue), with Jolly & Latifi (UNLV)

OUTLINE

Need, problem, solution

Need

Traffic growth, diverse applications, and what destination-based shortest-path routing leaves undone.

Problem

Formal definition, resource and constraint types, and why it is hard: tractability, stability, robustness, scalability.

Solution

CBR architecture, policy and QoS routing, algorithms, RSVP and DiffServ, MPLS with CR-LDP or RSVP-TE, and simulation results.

NEED

Traffic is growing and changing

Heavy load

Web, multimedia and corporate VPNs push heavy traffic through the Internet core. Bandwidth demand keeps outgrowing expansion.

Costly capacity

Providers must raise circuit bandwidth or router capacity, or add core routers. Existing capacity needs better use.

New demands

Multimedia needs a class of service per application. Service agreements between providers and users need new routing paradigms.

NEED

How routing works today

	Intra-domain (IGP)	Inter-domain (BGP)
Protocols	RIP, OSPF, IS-IS	BGP
Algorithm	Distance vector (RIP) or link state with Dijkstra (OSPF)	Path vector: the AS path is sent with each route
Decision	Fewest hops or lowest delay to the destination address	Policy-driven choice among advertised AS paths
Ignores	Jitter, congestion, load, per-flow requirements	Resource availability and QoS

Both forward on the destination address only. Source: Younis & Fahmy, Table I.

NEED

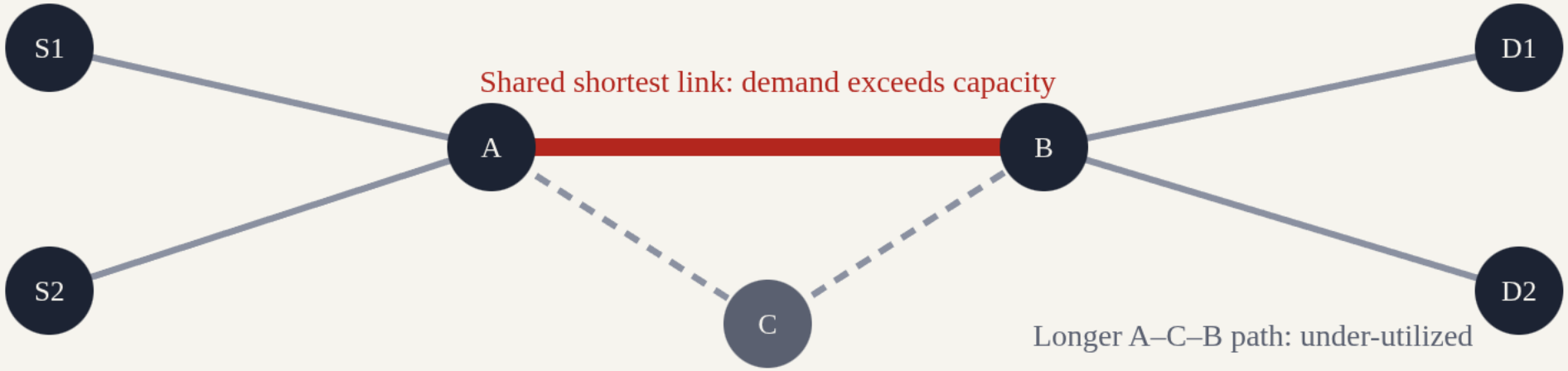
Different traffic needs different paths

Traffic	Requirement	Example constraint
Voice, video (RTSP)	Low delay, low jitter, bandwidth	End-to-end delay ≤ 40 ms
VPN	Isolation, reserved capacity	Only links assigned to that customer
Web, FTP	Best effort	No reservation
Gold customer	SLA-backed service	Route via the high-bandwidth path
Provider	Cost, load balance, security	Minimize cost

Constraint examples follow the paper's Figures 1, 7 and 9.

NEED

Shortest paths overload some links and idle others



The paper's point: shortest paths from different sources overlap on the same links while a longer path between the same routers is idle.

PROBLEM

Formal definition of the CBR problem

Network

Graph G = (V, E). V: routers or end systems. E: links. Each link j carries a value r(i,j) for each resource i, such as bandwidth, delay and loss.

Constraints

C = (c1, …, cn), where ci is the constraint on resource i. Policy constraints come from administrators, QoS constraints from applications.

Goal

Find a path p from source to destination such that every ci is satisfied, while reducing cost, balancing load or increasing security.

Figure 6 of the paper uses n = 3 resources (bandwidth, delay, loss) on 8 links.

PROBLEM

Resource and constraint types

Resource	Class	Combines along a path	Example
Available bandwidth	Concave	Minimum over links (bottleneck)	Path of 10 Mbps and 1.5 Mbps links offers 1.5 Mbps
Delay, jitter, cost, hop count	Additive	Sum over links	Delays of 12 + 25 ms give 37 ms
Loss probability	Multiplicative	Product of per-link success	Reserved bandwidth keeps it low
Boolean constraint	Feasibility	Pass or fail	Delay ≤ 40 ms
Quantitative constraint	Optimization	Ranks feasible paths	Highest bottleneck bandwidth

Resources are also configurable (propagation delay), dynamic (available bandwidth) or topological (path length). Table V gives the class of each resource. The delay split in the example row is illustrative.

PROBLEM

Why it is hard: tractability

NP-complete

With independent, real-valued resources, satisfying two boolean constraints, or one boolean and one optimization constraint, is NP-complete.

Polynomial cases

If all resources but one take bounded integers, or resources are dependent, the problem is solvable in polynomial time.

Practical cost

Hop-by-hop approaches run in linear time. Source routing needs path computation of about $O(|V| \log |V| + |E|)$, as in Wang-Crowcroft and Guerin-Orda.

Most proposals therefore use simple heuristics and keep global state.

PROBLEM

Stability, robustness and scalability

Stability

Frequent link updates cost bandwidth; infrequent ones leave stale state. Updates fire on absolute or relative change, and on timers with a minimum spacing. Dynamic routing risks route flapping.

Robustness

State is inaccurate because of update limits, few reporting nodes and aggregation. Bandwidth-only requests tolerate it; delay requirements do not. Links also fail or congest.

Scalability

Costs are computation (path criteria, triggers, alternate paths) and protocol overhead (update frequency and size). The OSPF QoS study found updates took under 1% of bandwidth.

SOLUTION

CBR architecture: where it sits

Management interface: trunk attributes, traffic, resource and policy specifications

Conventional intra-domain routing

OSPF-style process. Reads and updates the link state database. Still computes default shortest paths.

MPLS constraint-based routing process

Runs in layer 3 on each router. Computes feasible paths, exchanging state with other CBR processes and the IGP.

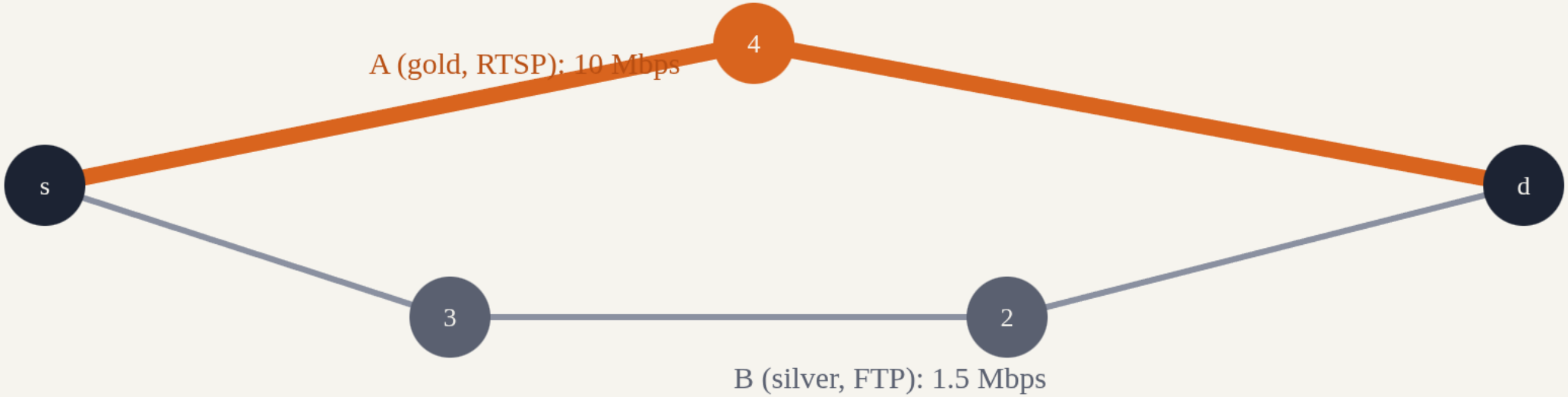
Resource attribute availability database

Holds the bandwidth, delay and policy data that CBR decisions need, carried by routing protocol extensions.

Off-line CBR computes paths outside the network from a known traffic matrix. Online CBR computes at routers and adapts to changes. Younis & Fahmy, Fig. 3 and Table II.

SOLUTION

Policy routing: example



Policy: “RTSP traffic should be routed through router 4.” Flow A takes s → 4 → d. Flow B stays on the default s → 3 → 2 → d, which is enough for best-effort FTP. In inter-domain routing each node is an AS, and rules like “A’s traffic to AS 4” at s and “A’s traffic to s” at AS 4 create a loop. This is a persistent route oscillation.

SOLUTION

Policy routing: implementation

Policy framework

Console → LDAP policy repository → PDP (policy server) → COPS → PEPs (routers). COPS exchanges rules; LDAP retrieves them.

Cisco IOS

CEF forwards from a FIB instead of a route cache. dCEF distributes it. NetFlow adds accounting and policy routing on top.

Linux and BGP

Linux has policy routing. BGP uses E-BGP (between ASes) and I-BGP (within one). Path asymmetry in I-BGP can cause divergence and path deflection.

Remaining gaps: policy consistency across repositories, authentication, traffic identification, mobile clients and common standards. Fix for conflicts: fewer manual settings and routing policy registries.

SOLUTION

QoS routing: path selection

Worked example

Constraints C = (delay ≤ 40 ms, minimize cost). The selected path is s → 3 → 2 → d with cost 12 and delay 37 ms. With “cost ≤ 13”, s → 1 → 2 → d is also viable.

Widest-shortest

Among the minimum-hop paths, choose the one with the greatest bottleneck bandwidth. Gives basic load balancing.

Shortest-widest

Among the paths with the highest bandwidth, choose the one with the fewest hops.

Order of the two optimization constraints matters. Pinned routes keep a flow on its selected path.

SOLUTION

QoS routing algorithms

Problem	Approach	Algorithm
Bandwidth-bounded	Source	Modified Bellman-Ford
Delay-bounded	Hop-by-hop	Split end-to-end delay into per-link constraints
Bandwidth- and delay-bounded	Source	Prune links lacking bandwidth, then Dijkstra on delay
Bandwidth- and cost-bounded	Source	Extended Bellman-Ford or Dijkstra, polynomial time
Delay-bounded, cost-optimized	Source	LARAC (Lagrange relaxation)
Multi-constrained	Source	H_MCOP heuristic
Multicast	Source	Constrained Steiner trees; MOSPF, QMRP, DVMA

Younis & Fahmy, Tables VI and VII.

SOLUTION

Reserving resources and keeping the path

RSVP and IntServ

CBR only finds a path. RSVP signals along it: admission and policy control, then a per-flow QoS table consulted by the classifier and scheduler. If the route changes, the old reservation times out and is re-set up, then the flow moves.

DiffServ

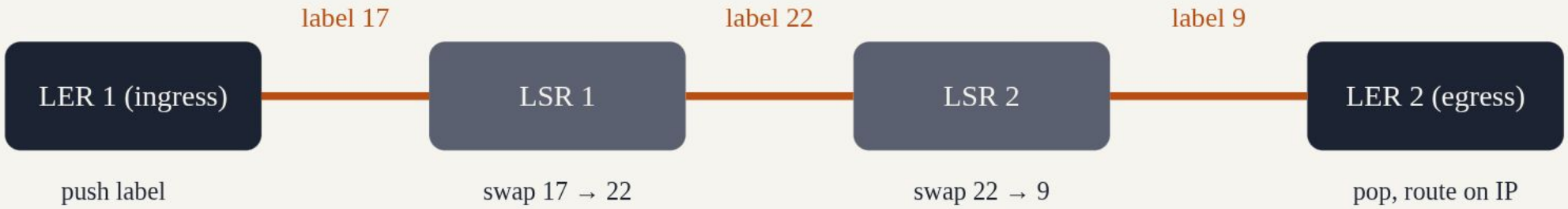
Edge routers mark a 6-bit DSCP in the IP header according to SLAs between domains. Core routers pick drop and scheduling behavior from it. It resembles an MPLS label but cannot prevent congestion without provisioning.

RSVP-TE

Runs on routers and takes the explicit route from a CBR algorithm. It carries that route in its messages and installs labels along the path, so LSPs avoid bottlenecks.

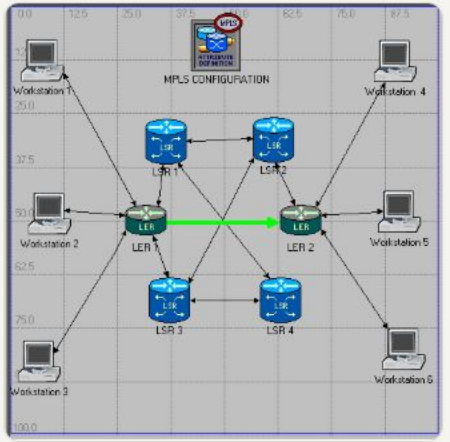
SOLUTION

MPLS: label swapping



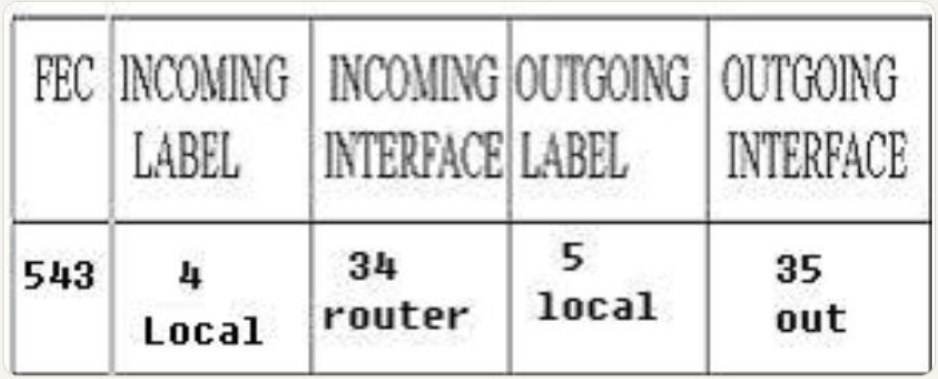
Label values are illustrative. Each label has only local meaning.

- The ingress LER classifies a packet once into a FEC and chooses the LSP.
- Each LSR looks up the incoming label, swaps it and forwards, using a table in hardware.
- The egress LER removes the label and forwards by IP.



SOLUTION

MPLS header and label information base



FEC	INCOMING LABEL	INCOMING INTERFACE	OUTGOING LABEL	OUTGOING INTERFACE
543	4 Local	34 router	5 local	35 out

- **Label**, 20 bits: switching value
- **CoS**, 3 bits: queuing and discard
- **S**, 1 bit: label stack
- **TTL**, 8 bits

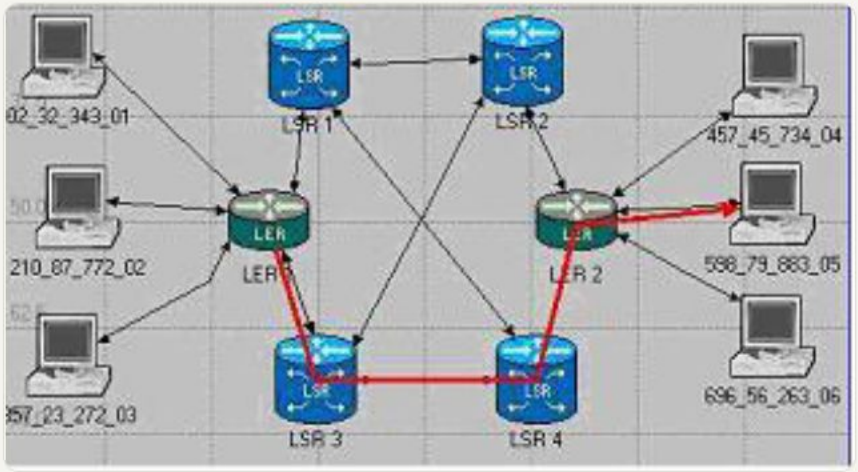
FEC	In label	In interface	Out label	Out interface
543	4	34	5	35

The 32-bit header sits between the layer 2 and IP headers. Where layer 2 has a label field (ATM VPI/VCI, Frame Relay DLCI) the label goes there. Example LIB row from Jolly & Latifi, Table 1.

SOLUTION

Setting up an LSP

- Control-driven (hop-by-hop or LDP) or explicit route (ER-LSP).
- An explicit route lists the hops from ingress to egress. Strict or loose.
- CR-LDP: label requests go downstream hop by hop, label mappings come back upstream.
- Downstream-on-demand: request a binding. Downstream-unsolicited: advertise without request.
- Retention: conservative keeps only valid next-hop labels. Liberal keeps all for faster reroute.



CR-LDP uses TCP for control and UDP for peer discovery. Figure from Jolly & Latifi, Fig. 5.

SOLUTION

CR-LDP versus RSVP-TE

	CR-LDP	RSVP-TE
Basis	Extended LDP	Extended RSVP
Signaling	TCP control, UDP discovery	IP datagrams or UDP
State	Hard	Soft, periodic refresh
LSP setup	Forward, then upstream mapping	Backwards, upstream
Failure recovery	Local and global	Local and global
Scalability	Good	Marginal
Multicast LSP	No	No (RSVP: yes)

Both support strict and loose routes, pinning and loop prevention. Jolly & Latifi, comparison table.

SOLUTION

Simulation set-up (OPNET)

- Topology: 8 label switch routers and 5 edge routers across the United States.
- One LSP from Los Angeles to New York.
- The same scenario runs once with CR-LDP and once with RSVP.
- Measured: LSP traffic in, LSP setup time, LSP delay.
- Simulation ran 45 minutes.

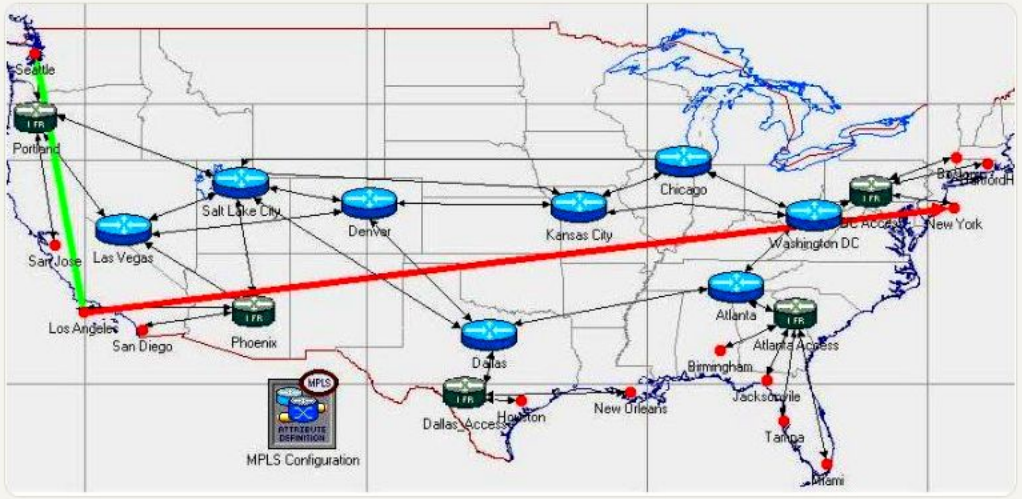


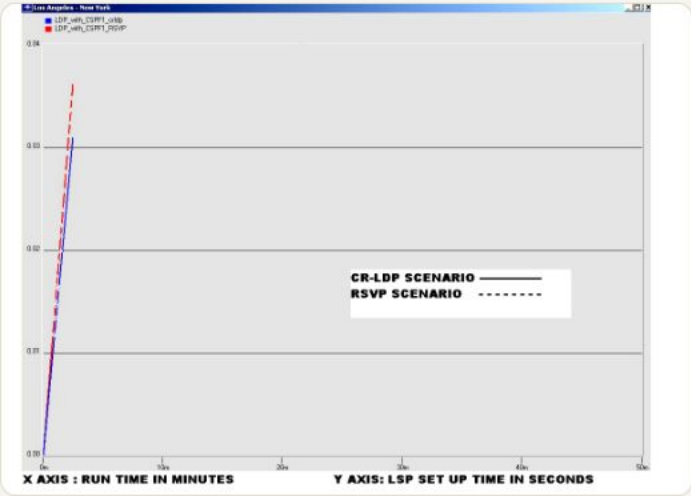
Figure 6 from Jolly & Latifi.

SOLUTION

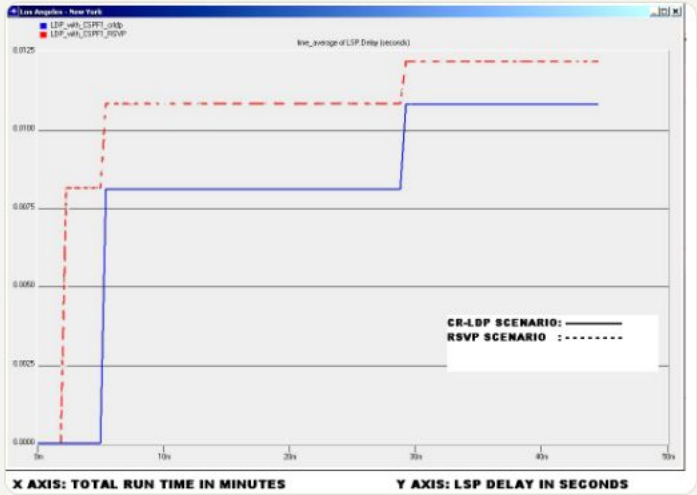
Simulation results



Traffic: curves nearly overlap, so throughput matches.



Setup time: CR-LDP sets up faster.



Delay: CR-LDP lower; the gap varies over time.

Single scenario from Jolly & Latifi, Figs. 7 to 9.

SOLUTION

Handling the hard parts

Stability

Update on significant change (absolute classes or relative threshold) plus timers. Route long-lived flows dynamically and short ones statically. Mix conventional and QoS routing.

Robustness

Safety routing estimates bandwidth probabilistically. Prune congested links. Proportional sticky routing spreads load using flow blocking probability.

Scalability

Topology aggregation, limiting decisions to edge routers, partitioning QoS requirements and ticket-based probing.

Large IP domains may instead tune OSPF weights, and MPLS global deployment remains uncertain because of cost and complexity.

TAKEAWAYS

What to remember

- Destination-only shortest-path routing ignores service needs and leaves capacity idle.
- CBR formalizes the choice: find a path meeting constraints on additive, concave and multiplicative resources.
- Policy comes first and QoS second; hard cases use heuristics.
- RSVP reserves, DiffServ marks, and MPLS with CR-LDP or RSVP-TE pins the chosen path.
- In simulation CR-LDP set up faster with lower delay; the paper expects the two to converge.
- Open issues: complexity, stability, inaccurate state, deployment cost.